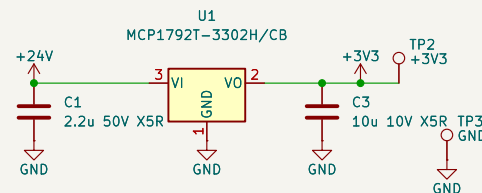
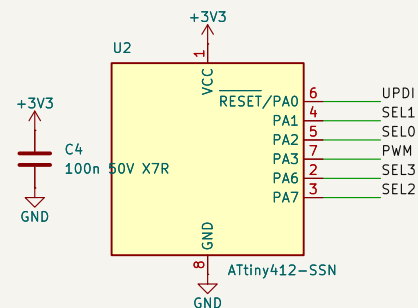


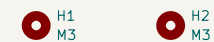
3.3 V regulator



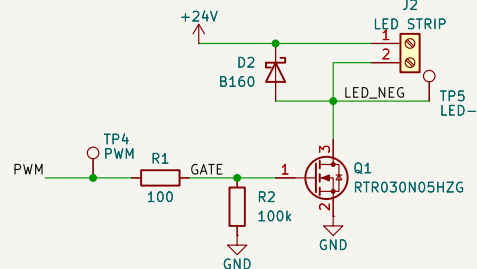
MCU



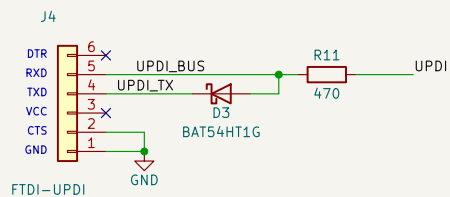
Mechanical



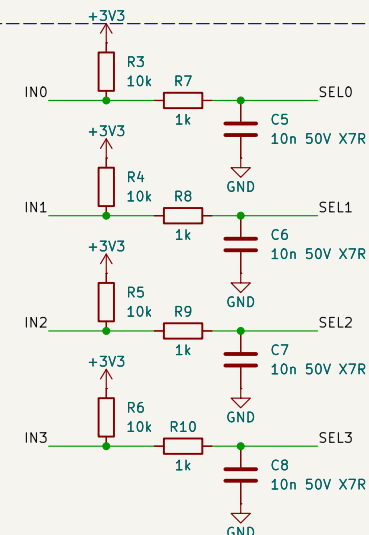
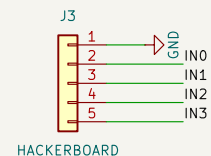
LED output stage



Programming (SerialUPDI)



Hackerboard inputs



Hackerboard pins 0–3 are open-drain: 1 = pulled to GND.

Firmware: level = $\sim(\text{SEL3}..\text{SEL0}) \& 0x0F$, so open inputs = level 0 (off).

PWM: TCA0 W00 on PA3, 10-bit, ~2.44 kHz. R2 holds GATE low in reset.

Flash: 3.3 V USB–serial adapter on J4, board powered from 24 V.

```
avrdude -c serialupdi -p t412 -P <port> -U flash:w:fw.hex
```

Level selected by Prusa GPIO Hackerboard pins 0–3
ATtiny412 + AO3422 low-side PWM for a 24 V LED strip

hyiger

Sheet: /

File: led-pwm-4bit.kicad_sch

Title: 4-Bit LED PWM Controller

Size: A4	Date: 2026-10-07
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Size: A4	
KiCad E.D.A. 10.0.7	

Rev: 0.1

Id: 1/1